

Cysts - Histology

OMFS 661: Diagnostic Oral Radiology and Pathology

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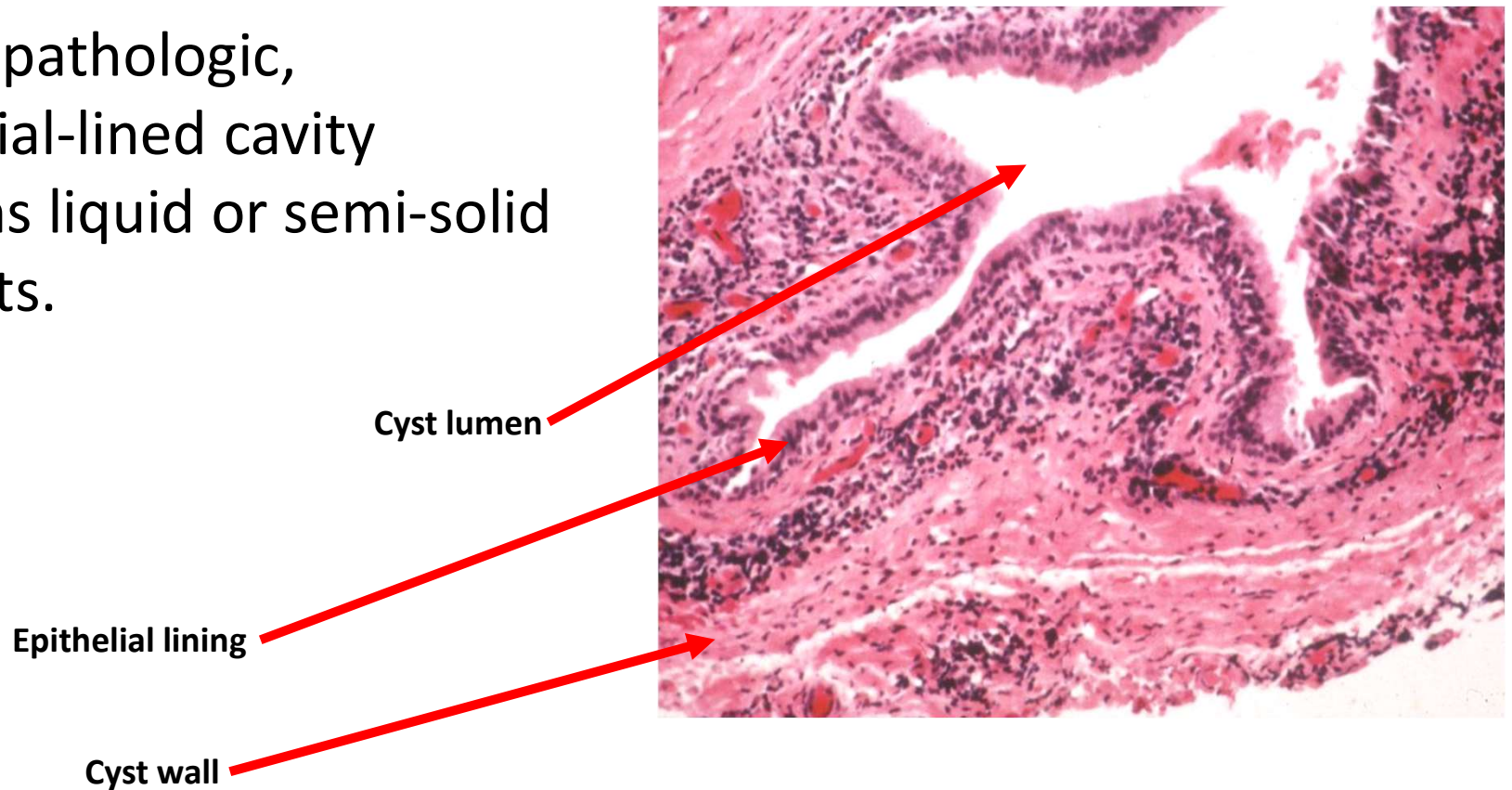


Mission: To advance the science and art of health through education, service, scholarship and social accountability

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Definition

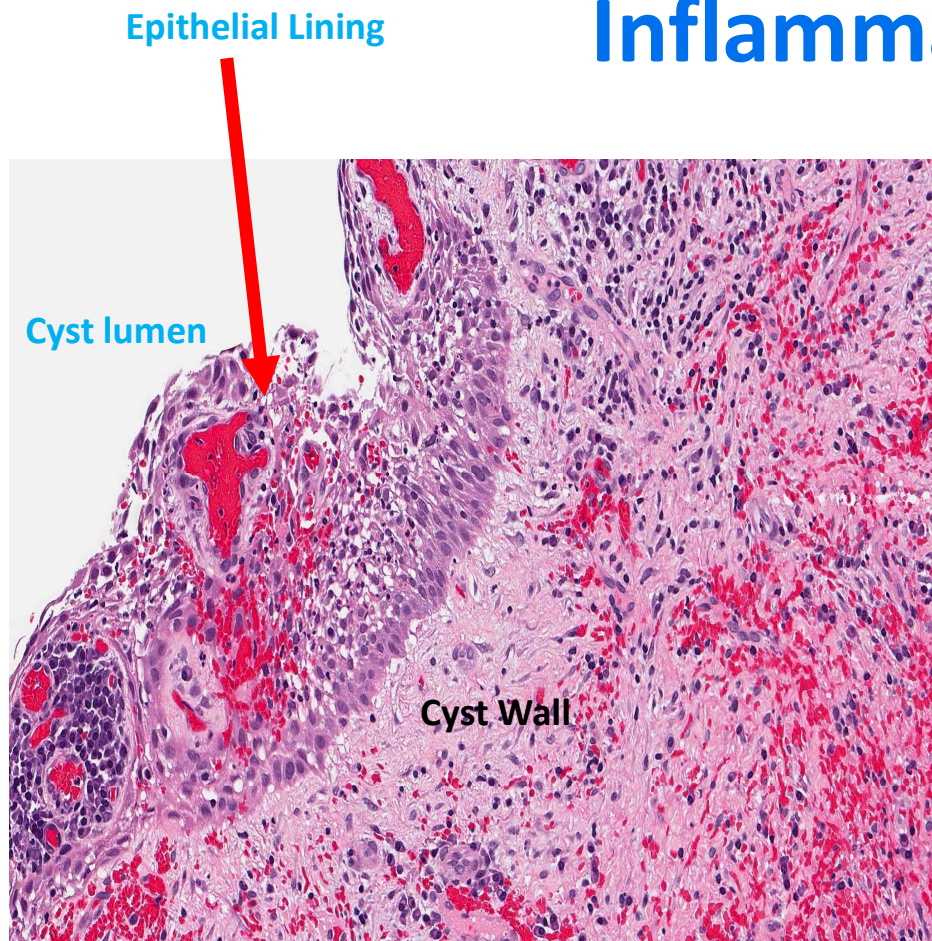
- Cyst- a pathologic, epithelial-lined cavity contains liquid or semi-solid contents.



Inflammatory Cysts

- True cysts arising from a non-vital tooth
 - Rest of Malassez (epithelial remnants derived from Hertwig's epithelial root sheath)
- **Periapical/Radicular cyst**
 - Located at the **apex** of the tooth
- **Lateral radicular cyst**
 - Located along the **lateral surface of the root**
- **Residual cyst**
 - Periapical cyst **located at the site where a non-vital tooth was once present**
 - **Epithelium may undergo malignant transformation**
 - **Squamous cell carcinoma (interosseous carcinoma)**

Inflammatory Cyst



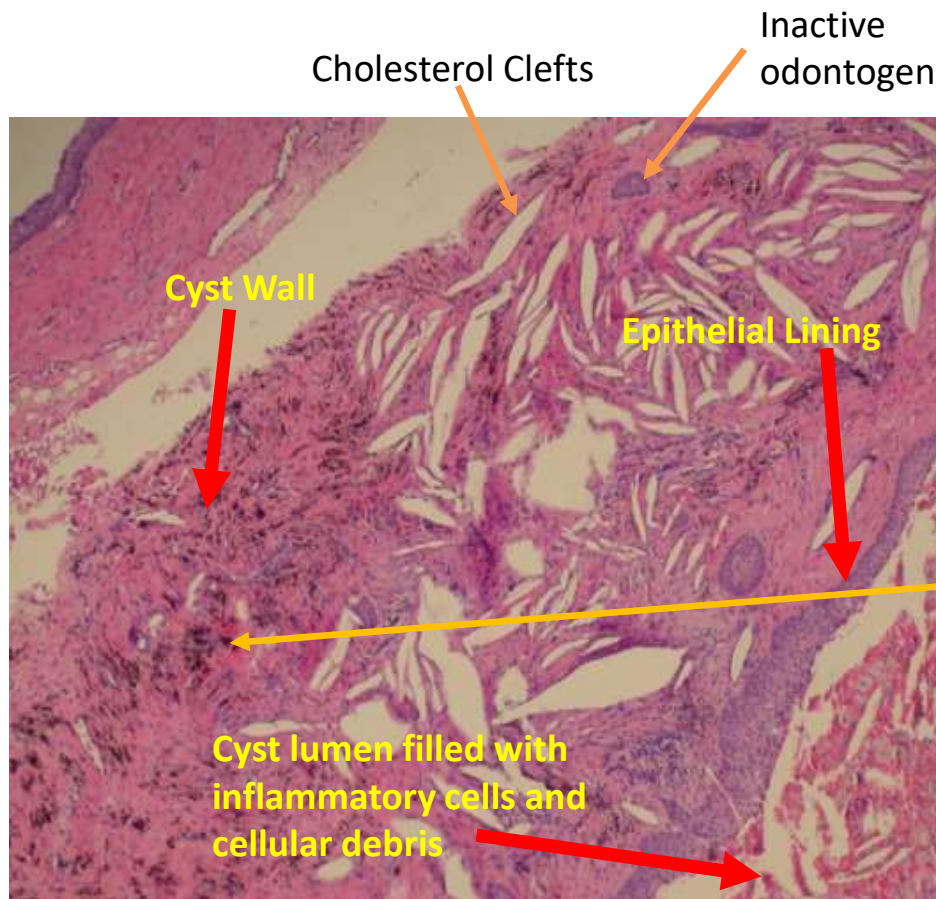
Histology

- **Chronic inflammation** of the cyst wall
 - **Lymphocytes and plasma cells**
 - Occasionally pockets of neutrophils can be seen (subacute inflammation).
- **Non-keratinized epithelium** lining the cyst wall
- **Cyst lumen**

Magliocca K. Radicular (periapical) cyst. PathologyOutlines.com website.
<https://www.pathologyoutlines.com/topic/mandiblemaxillaperiapicalcyst.html>

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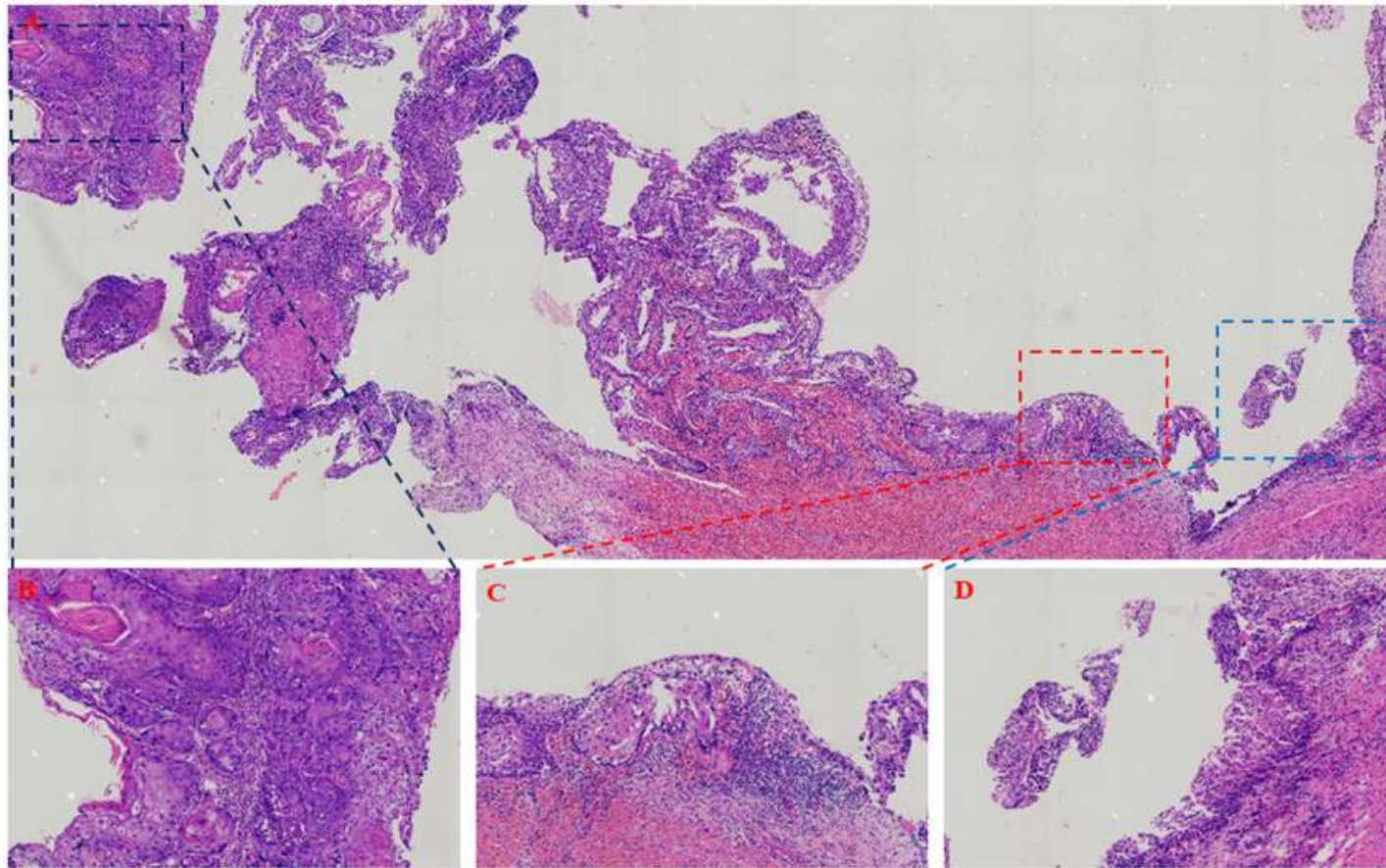
Inflammatory Cyst



- Histology

- **Chronic inflammation** of the cyst wall
 - **Lymphocytes and plasma cells**
 - Occasionally pockets of neutrophils can be seen (subacute inflammation)
 - Brown pigmentation, cholesterol cleft amongst other feature appear in the wall.
- **Non-keratinized epithelium** lining the cyst wall
- **Lumen** can be filled with fluid and sometimes cellular debris

Residual Cyst with Malignant Transformation



SCC present within the epithelium

Hyperplastic epithelial lining with keratin formation

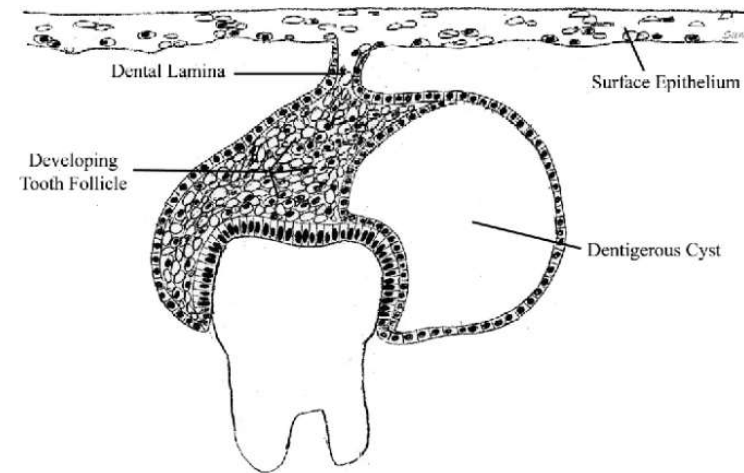
Non-keratinized epithelial lining

He, Zongxuan & Tao, Yueqin & Li, Jieying & Shang, Wei & Song, Kai. (2024). Primary intraosseous squamous cell carcinoma originating from a mandibular odontogenic cyst: A case report. Science Progress.

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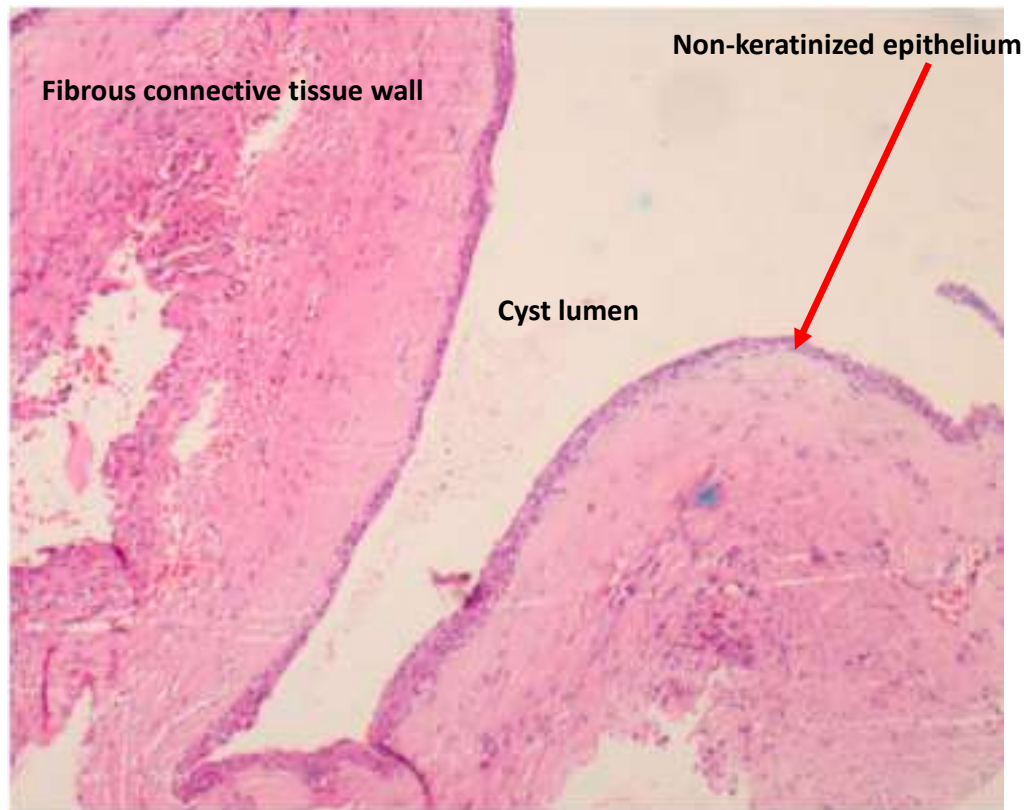
Dentigerous Cyst

- Aka follicular cyst
- The cyst develops from the remnants of the reduced enamel epithelium
 - Fluid accumulation builds between the crown of tooth and the reduced enamel epithelium
- The cyst lining is multipotent
 - The cells can differentiate into a limited number of cell types



McCoy, J. Michael. "Complications of retention: pathology associated with retained third molars." *Atlas of the oral and maxillofacial surgery clinics of North America* 20 2 (2012): 177-95 .

Dentigerous Cyst

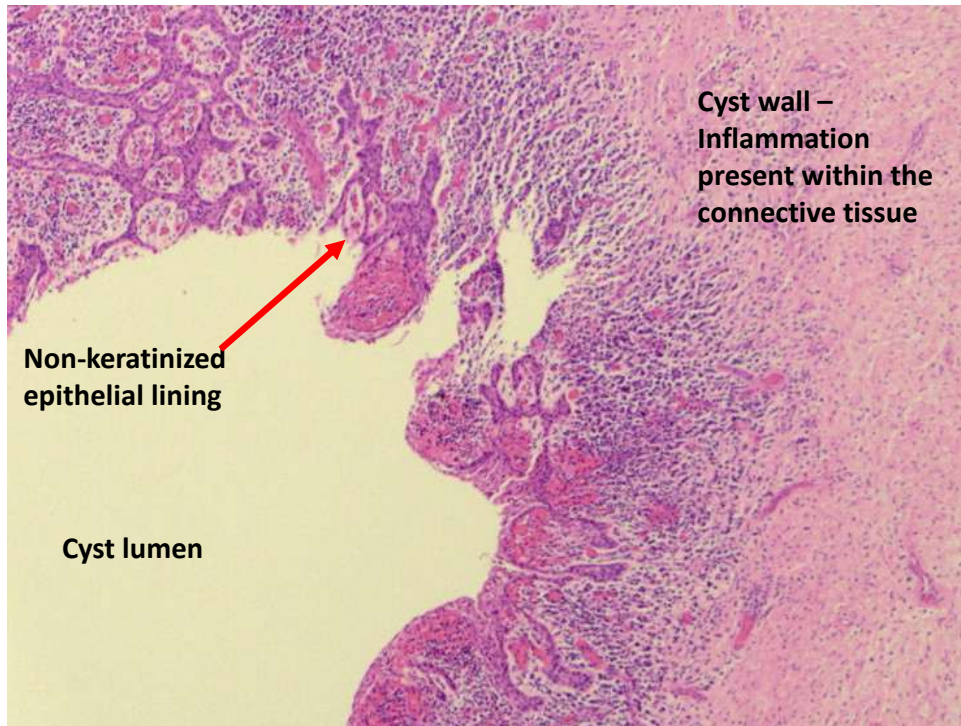


- Histology

- Thin layer of **non-keratinized epithelium/reduced enamel epithelium** lining the cyst wall
- Cyst wall is composed of **tissue resembling the primitive tissues of tooth structure.**
 - Inflammation can be seen
- Cyst lumen contains fluid and may also have cellular debris.

K.K. Deepa, et al. A case of dentigerous cyst in a pediatric patient - With an insight into differential diagnostic entities, *Advances in Oral and Maxillofacial Surgery*, Volume 3, 2021, 100130, 6/25/2026

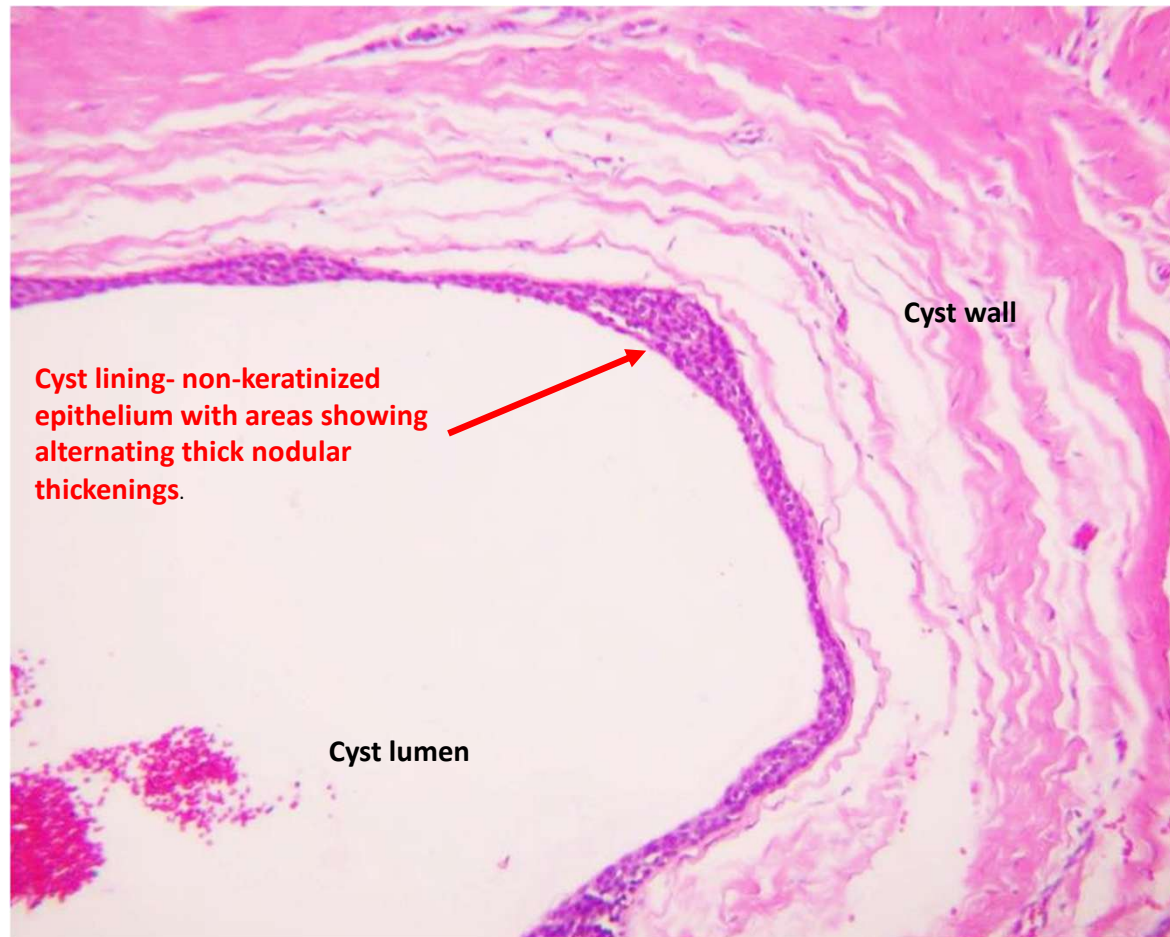
Buccal Bifurcation Cyst



- Histology
 - **Mimics the histology seen with dentigerous or radicular cysts**
 - Clinical correlation must be used to make the distinction between the two cysts.
 - Inflammation within the cyst wall can altered the appearance of the epithelial lining

Journal of Oral and Maxillofacial Surgery
Volume 72, Issue 10, October 2014, Pages 1966-1973

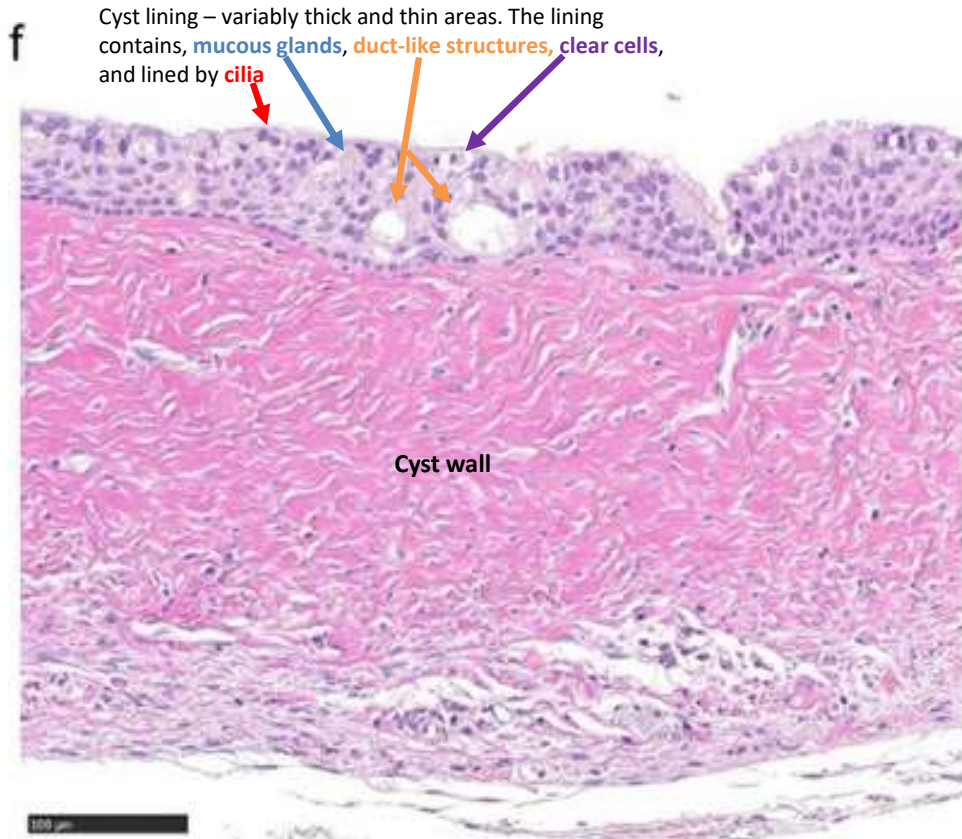
Lateral Periodontal Cyst



Histology

- Cyst wall
 - Usually devoid of inflammation
- Epithelial lining
 - **Alternating thin and thick (plaque-like) areas.**
- Cyst lumen
 - May contain fluid, cellular debris, or blood

Glandular Odontogenic Cyst

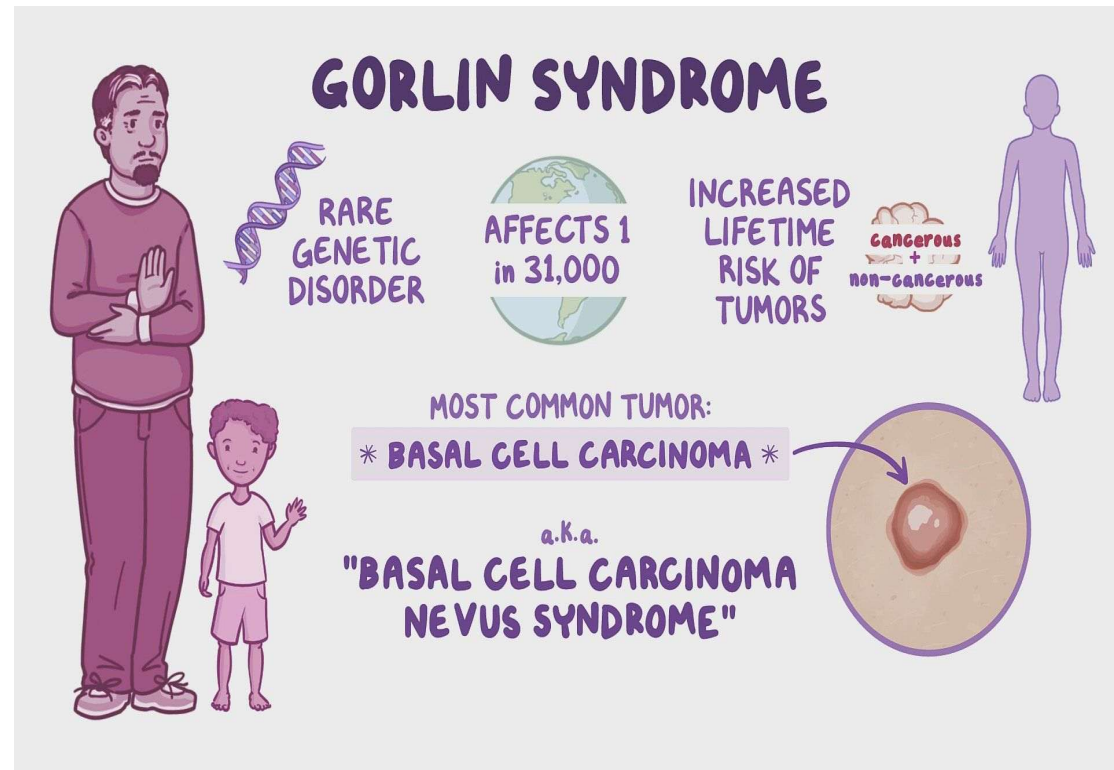


Histology

- Thick, epithelial lining with portions extending into the lumen
 - **Mucous cells and duct-like structures are seen with the lining**
 - **Cilia** can be seen on the surface

Odontogenic Keratocyst

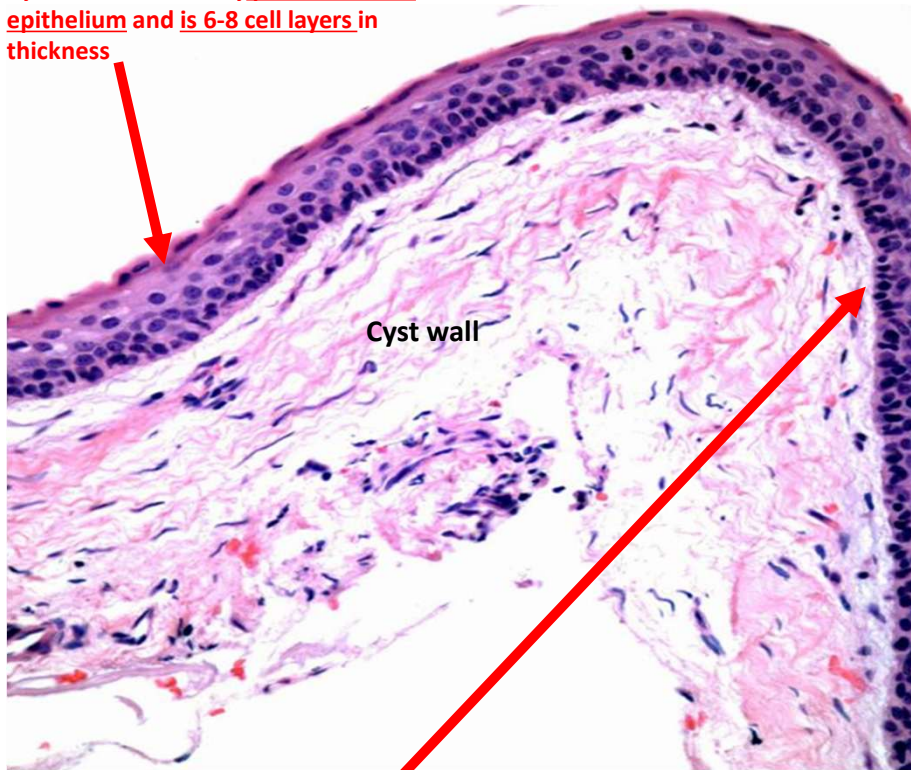
- Nevroid Basal Cell Carcinoma Syndrome
 - Gorlin Syndrome
- Cause- chromosomal mutation of the PTCH gene
 - 9p22.3-q31



Osmosis

Odontogenic Keratocyst

Cyst wall is formed by parakeratinized epithelium and is 6-8 cell layers in thickness



Palisading of basal cell layer due the vertical orientation "Tombstones"



Satellite cysts in the cyst wall

Odontogenic Keratocyst



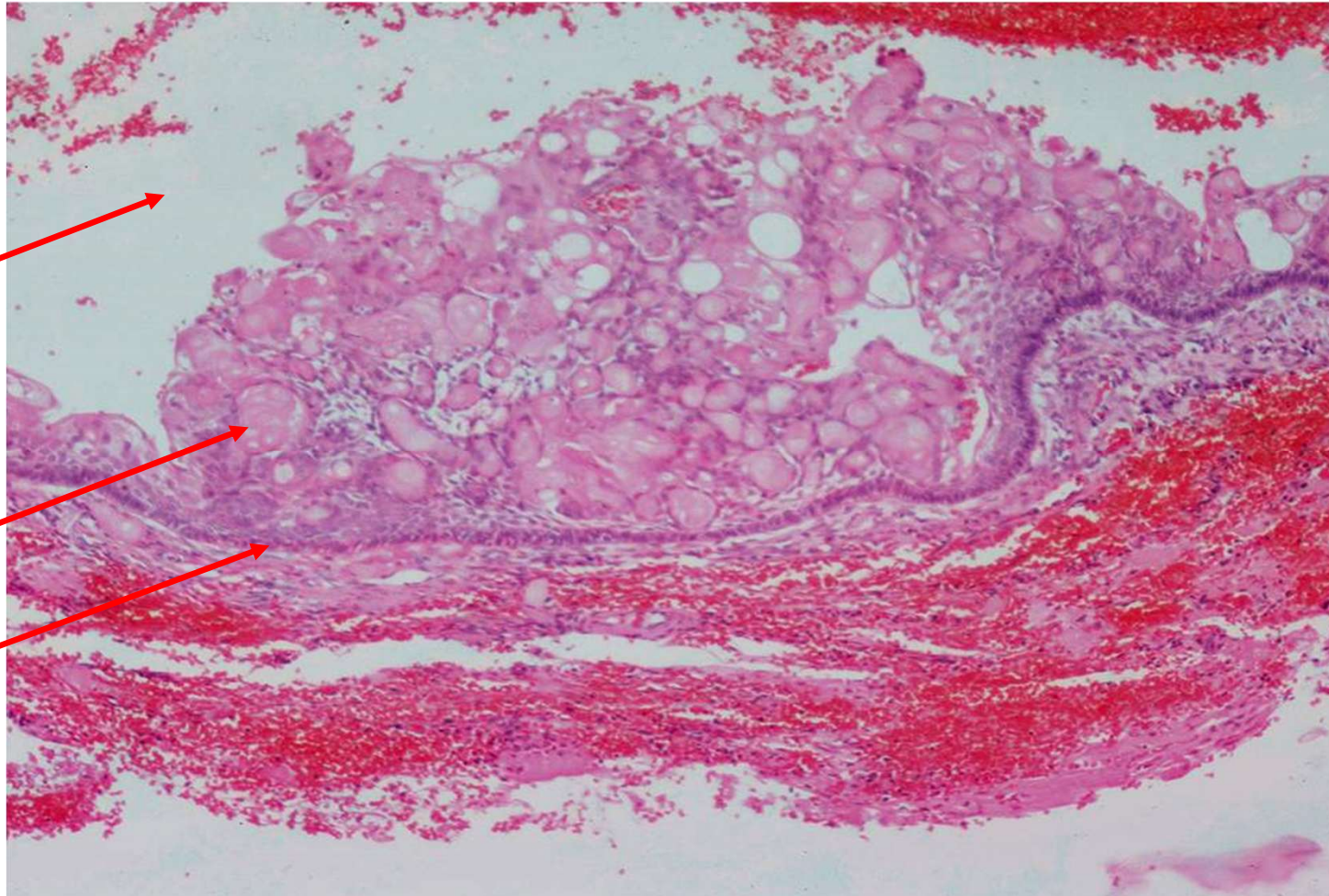
Corrugated, parakeratinized
epithelium with prominent
basal cells

Cyst lumen filled with
keratin scales

Calcifying Odontogenic Cyst (Gorlin Cyst)

- Epithelial lining
 - Tall basal cell layer (cuboidal or columnar cells)
 - Loose squamous cells noted superficially towards the lumen
- Key histologic feature
 - Pink cells missing the nuclei **"Ghost cells"**
- Aggressive variant – Dentinogenic Ghost Cell tumor

Calcifying Odontogenic Cyst



Cyst
lumen

Ghost cell
with missing
nuclei

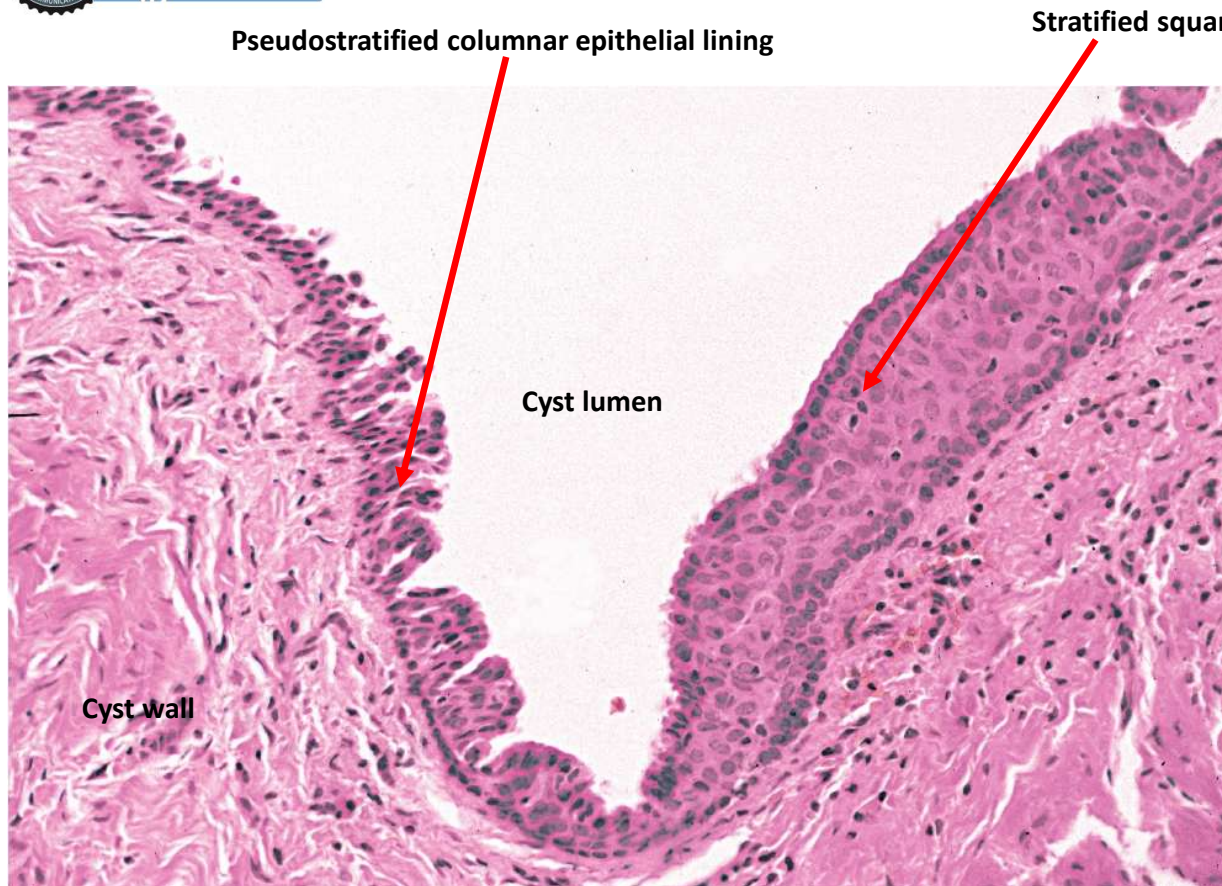
Epithelium
lining the
cyst wall

6/25/2026

Other Cysts

- Non-Odontogenic Developmental Cyst
 - Nasopalatine Duct Cyst (Incisive Canal Cyst)
- Pseudocysts
 - Simple Bone Cyst (Traumatic Bone Cyst)
 - Stafne's Bone Defect

Nasopalatine Duct Cyst

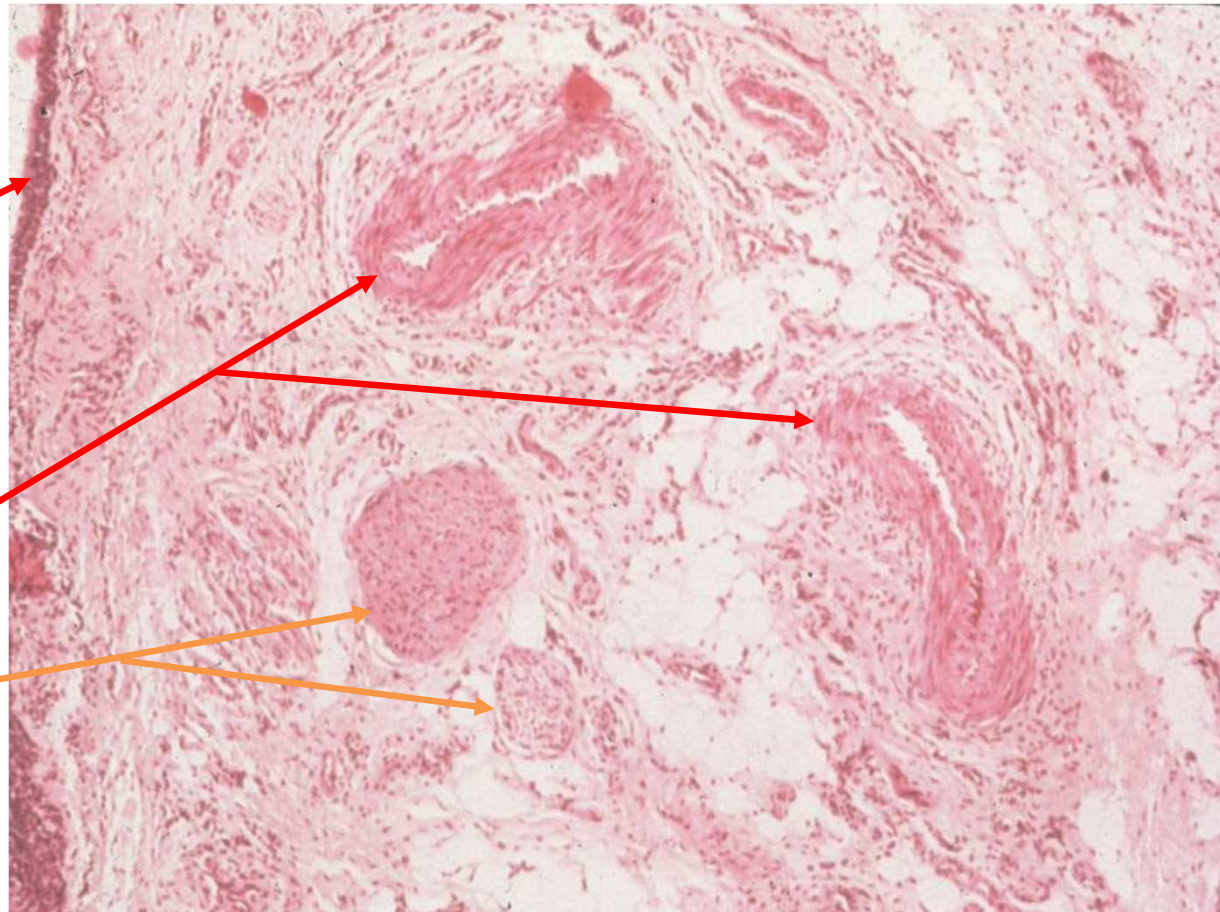


Histology

- Epithelial lining
 - Can be respiratory in appearance
 - **Ciliated**
- Cyst wall containing the contents of the incisive canal
 - **Nerve and blood vessels**
 - Glandular tissue

Oral and Maxillofacial Pathology 5th Edition Neville

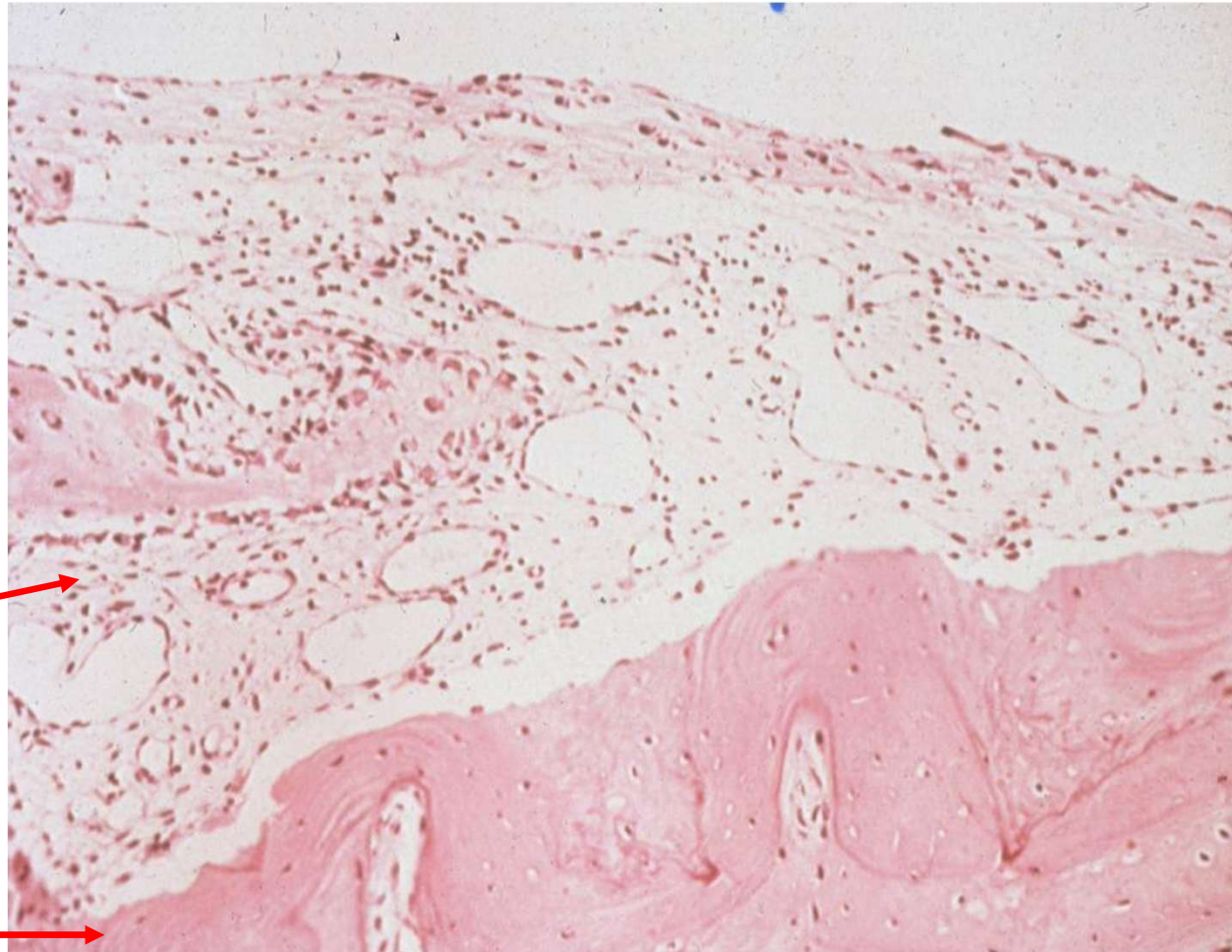
Nasopalatine Duct Cyst



Cyst lining which has features of the nasal epithelium (cilia are noted on the epithelial surface)

Cyst wall containing **blood vessels** and **nerves bundles**

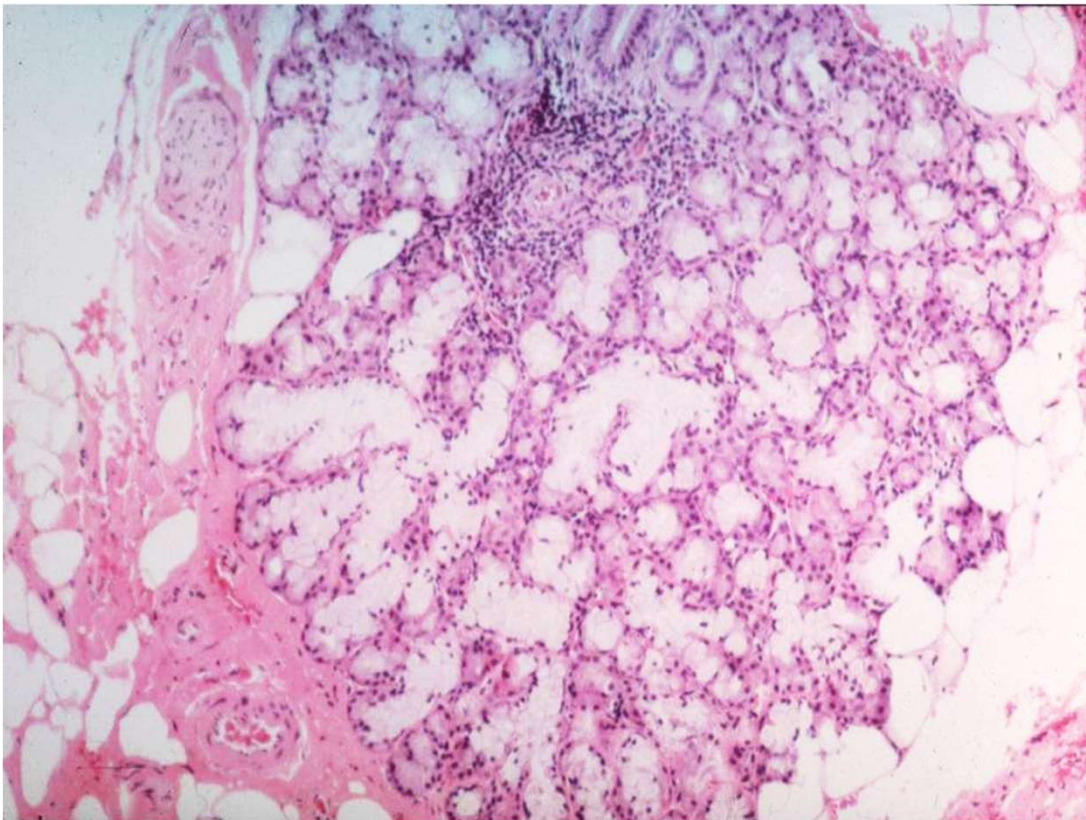
Simple Bone Cyst



Connective tissue
wall with blood
vessels and **NO**
epithelial lining

Vital bone can
be seen in the

Stafne's Bone Defect



Histology shows: a combination of normal salivary gland, fat, and connective tissue with blood vessels and nerves (**normal tissue**)

Resources

- Chapter 1: Nasopalatine Duct Cyst (Developmental Cyst) & Stafne's Bone Defect (Pseudocyst)
- Chapter 3: Inflammatory Cysts
- Chapter 14: Simple Bone Cyst (Pseudocyst)
- Chapter 15: Odontogenic Cysts

